

Humanetics Innovative Solutions, Inc. v. Kistler Instrument

A public complaint analysis describes allegations that Kistler's adjustable THOR-50M spine-box and WS-50M ankle assemblies infringe patents concerning an adjustable lumbar-spine mechanism and an ankle mechanism with adjustable friction and elastomeric elements for inversion and eversion.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-25	U.S. District Court for the Eastern District of Michigan	2:26-cv-13026	Patent Infringement

Prepared 2026-08-27. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

Public filing, likely recipient, and technical frame

CAPTION	Humanetics Innovative Solutions, Inc. v. Kistler Instrument Corporation et al.
FORUM / DOCKET	U.S. District Court for the Eastern District of Michigan / 2:26-cv-13026
FILED / TYPE	2026-08-25 / patent infringement
PUBLIC DOCKET	Open docket

VERIFIED PUBLIC RECORD

The public docket identifies Humanetics as plaintiff, Kistler Instrument Corporation and Kistler Instrumente AG as defendants, the August 25, 2026 filing date, docket number 2:26-cv-13026, and Kristopher Hulliberger as the filer. It lists a patent complaint under 35 U.S.C. § 271 with jury demand and identifies exhibits for U.S. Patent Nos. 9,965,977 and 8,622,748 plus exemplary claim charts.

LIKELY RESEARCH PURCHASER

Kristopher K. Hulliberger - Outside counsel, Howard & Howard Attorneys PLLC. The current public docket names Hulliberger as filer of Humanetics' complaint. Treating the opening filer as a potential purchaser is an analytical inference, not a verified purchasing decision.

khulliberger@howardandhoward.com - publicly printed in the [linked professional source](#); not inferred.

TECHNICAL FRAME

A public complaint analysis describes allegations that Kistler's adjustable THOR-50M spine-box and WS-50M ankle assemblies infringe patents concerning an adjustable lumbar-spine mechanism and an ankle mechanism with adjustable friction and elastomeric elements for inversion and eversion.

Secondary-source limitation: This is preliminary public-source analysis. The filed complaint, its exhibits and claim charts, product inspection, source design records, testing, and discovery must control any final opinion. [Open public synopsis](#)

REPORTED ASSERTED PATENTS

[U.S. Patent No. 9,965,977](#) | [U.S. Patent No. 8,622,748](#)

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 Adjustable lumbar-spine assembly

Does the accused THOR-50M spine box include the claimed upper and lower members, abdomen interface, and adjustment mechanism, including adjustment at different increments about a lateral axis?

Potential evidence: Physical inspection, CAD and drawings, bills of materials, technical bulletins, angle/position testing, product versions, and expert measurement.

02 Biomechanical biofidelity of spine positioning

What occupant posture, load path, range of motion, and repeatability does each spine assembly produce, and do the accused adjustments perform the claimed biomechanical function?

Potential evidence: ATD specifications, test protocols, high-speed video, load-cell and displacement data, calibration records, and finite-element or multibody models.

03 Ankle clevis/base construction

Does the accused WS-50M ankle have the claimed clevis and base arrangement, fastener, and elastomeric elements in the claimed relationship?

Potential evidence: Component drawings, teardown/inspection, assembly instructions, parts lists, microscopy or materials testing, and dimensional analysis.

04 Friction adjustment and bracing simulation

Does the accused ankle's fastener permit friction adjustment that simulates human bracing in impact conditions, rather than merely joining components?

Potential evidence: Torque and friction testing, test protocols, design requirements, engineering notebooks, calibration settings, and impact-test data.

Questions likely to require expert input

05 Inversion and eversion response

Do the claimed and accused elastomeric elements provide the relevant inversion/eversion performance under representative crash-test loads?

Potential evidence: Material specifications, force-displacement curves, dynamic tests, data-acquisition traces, validation reports, and comparative biomechanical literature.

06 Product/version attribution and damages nexus

Which Kistler components, revisions, sales, installations, and instructions are implicated, and what technical contribution of the alleged features is supported by evidence?

Potential evidence: Revision histories, service bulletins, sales/installation records, test reports, customer records, product documentation, and technical-nexus analysis.

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 Jason R. Kerrigan

Commonwealth Professor of Mechanical and Aerospace Engineering and Director, Center for Applied Biomechanics, University of Virginia

TECHNICAL FIT

Direct fit for crash-test-dummy development and assessment, injury biomechanics, human-body computational models, and vehicle safety.

RELEVANT WORK

UVA describes his work on lumbar spine and pelvis response, crash-test-dummy injury prediction, human-body computational modeling, structural crashworthiness, and injury-risk mitigation.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Current director of a major injury-biomechanics center; potential institutional, sponsor, or industry relationships require diligence.

[Source 1](#)

[Draft email](#) | jason.kerrigan@virginia.edu | [public email source](#)

02 Jeff R. Crandall

Professor Emeritus, University of Virginia Center for Applied Biomechanics

TECHNICAL FIT

Direct fit for ATD development, biofidelity, crash injury biomechanics, and occupant-response testing.

RELEVANT WORK

UVA states that he contributed to development and evaluation of THOR and other advanced dummies, with experimental biomechanics from tissue to whole-body testing and more than 500 technical papers.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Emeritus status and extensive automotive-industry relationships warrant availability and conflict diligence.

[Source 1](#) | [Source 2](#)

[Draft email](#) | jrc2h@virginia.edu | [public email source](#)

Candidates 3-4 for further diligence

03 Matthew P. Reed

Research Professor, University of Michigan Transportation Institute

TECHNICAL FIT

Occupant accommodation, ATD positioning, anthropometry, posture, and test-device use in vehicle-safety research.

RELEVANT WORK

UMTRI identifies him as a researcher and SAE Human Accommodation and Design Devices Committee member, with transportation-safety research and long-standing SAE technical recognition.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Best aligned with occupant posture and ATD use; another candidate may be needed for detailed elastomer/friction component analysis.

[Source 1](#)

[Draft email](#) | mreed@umich.edu | [public email source](#)

04 Jason L. Forman

Research faculty, University of Virginia Center for Applied Biomechanics

TECHNICAL FIT

ATD response, occupant kinematics, impact biomechanics, and testing of seating/posture-related crash conditions.

RELEVANT WORK

A public UVA workshop identifies Forman and his UVA address; published proceedings identify him with the UVA Center for Applied Biomechanics. His work is directly relevant to THOR testing and ATD occupant-response questions.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Publicly identified source here is a workshop/proceedings record rather than a current faculty biography; current role and conflicts need confirmation before retention.

[Source 1](#) | [Source 2](#)

[Draft email](#) | jlf3m@virginia.edu | [public email source](#)

Candidates 5-6 for further diligence

05 Richard W. Kent

Professor, Mechanical and Aerospace Engineering, University of Virginia

TECHNICAL FIT

Injury biomechanics, tissue response, impact testing, computational mechanics, and vehicle crashworthiness.

RELEVANT WORK

UVA states that he leads its Automobile Safety Research Group and studies injury tolerance, loading conditions, and thorax structural behavior; a UVA account identifies his virtual-dummy work on thorax and upper-extremity modeling.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Focus is broader injury biomechanics, not necessarily the exact THOR-50M spine or WS-50M component architecture.

[Source 1](#) | [Source 2](#)

[Draft email](#) | richkent@virginia.edu | [public email source](#)

06 F. Scott Gayzik

Professor of Biomedical Engineering, Wake Forest University School of Medicine

TECHNICAL FIT

Computational injury biomechanics, virtual human-body models, ATD biofidelity, risk curves, and automotive crash testing.

RELEVANT WORK

Wake Forest describes his work developing realistic virtual body models for automotive crash tests and participation in ATD biofidelity-response corridors and injury-risk curve work for an anthropomorphic test device.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Computational modeling focus may require pairing with a hands-on ATD component/test specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | sgayzik@wakehealth.edu | [public email source](#)

Candidates 7-8 for further diligence

07 Narayan Yoganandan

Professor and Chair of Biomechanics, Medical College of Wisconsin

TECHNICAL FIT

Impact trauma, spine biomechanics, computational and experimental methods, and crash-test lumbar-spine loading.

RELEVANT WORK

MCW describes his research in impact trauma and clinical spine biomechanics; its publication record includes Hybrid III manikin lumbar-spine loading and dynamic lumbar-spine injury work.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

A medical/biomechanics orientation may need a mechanical-ATD specialist for claims focused on particular assembly geometry.

[Source 1](#) | [Source 2](#)

[Draft email](#) | yoga@mcw.edu | [public email source](#)

08 Frank A. Pintar

Professor, Marquette University and Medical College of Wisconsin Joint Department of Biomedical Engineering; Director, Wisconsin CIREN Center

TECHNICAL FIT

Motor-vehicle crash trauma, spine biomechanics, finite-element analysis, occupant injury, and test-method validation.

RELEVANT WORK

His institutional profile identifies research on blunt and penetrating trauma, occupant injuries, human impact tolerance, and NHTSA-funded CIREN work; MCW describes extensive motor-vehicle crash trauma and biomechanical research.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

May be most useful for function, biofidelity, and testing rather than claim-language construction.

[Source 1](#) | [Source 2](#)

[Draft email](#) | fpintar@mcw.edu | [public email source](#)

Candidates 9-10 for further diligence

09 Brian Stemper

Professor, Marquette University and Medical College of Wisconsin Joint Department of Biomedical Engineering; Director, Neuroscience and Biomechanics Research Laboratories

TECHNICAL FIT

Experimental injury biomechanics, spine and brain response, vehicle crashworthiness laboratory methods, and high-rate test/data-acquisition work.

RELEVANT WORK

His institutional profile identifies trauma-biomechanics teaching and the director role; MCW's core description identifies its vehicle crashworthiness laboratory, servo-sled, high-speed video, 3D motion capture, and data-acquisition capabilities.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

The public profile emphasizes injury response more than design of the two asserted mechanical components.

[Source 1](#) | [Source 2](#)

[Draft email](#) | stemp@mcw.edu | [public email source](#)

10 John H. Bolte IV

Professor, Health and Rehabilitation Sciences, The Ohio State University; Director, Injury Biomechanics Research Center

TECHNICAL FIT

Injury biomechanics, ATD instrumentation, impact testing, and automotive safety modeling.

RELEVANT WORK

A DOT report identifies Bolte as project lead and director of the Injury Biomechanics Research Center for work connecting bioinjury data, crash scenarios, and safety modeling; public NHTSA proceedings identify his ATD/injury-biomechanics work.

PUBLIC LITIGATION EXPERIENCE

No specific public litigation engagement is asserted from this bounded review.

POINTS FOR FURTHER DILIGENCE

Institutional email and affiliation should be rechecked before outreach because the cited DOT sources are older than the current matter.

[Source 1](#) | [Source 2](#)

[Draft email](#) | john.bolte@osumc.edu | [public email source](#)

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.
- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.
- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.
- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

[Schedule a 15-minute conversation](#)

CORE SOURCES

Public docket: <https://dockets.justia.com/docket/michigan/miedce/2%3A2026cv13026/397311>

Public professional email source: <https://howardandhoward.com/attorneys/kristopher-k-hullibarger>

Public technical context source: [https://ai-lab.exparte.com/case/dct/mied/2%3A26-cv-13026/doc/analysis/1](https://ai-lab.experte.com/case/dct/mied/2%3A26-cv-13026/doc/analysis/1)

LIMITATIONS

This is preliminary public-source analysis. The filed complaint, its exhibits and claim charts, product inspection, source design records, testing, and discovery must control any final opinion.

No attorney or expert was contacted. Availability, interest, compensation, and conflicts were not checked.